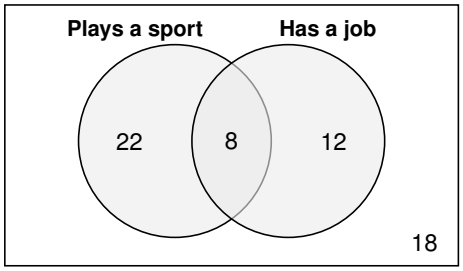
# Two-Way Tables and Relative Frequency

Name _____

60 students answered two questions: *Do you play a sport? Do you have a job?* A **two-way table** shows the answers to both questions at once.

| | **Job** | **No job** | **Total** |
|---|---|---|---|
| **Sport** | 8 | 22 | 30 |
| **No sport** | 12 | 18 | 30 |
| **Total** | 20 | 40 | 60 |




Each **cell** counts students who match **both** labels. 8 students play a sport **and** have a job.
The **totals** are in the margins. 30 students play a sport. 20 have a job.
The **grand total** is the corner. 60 students in all.

**Check:** Each row adds to its total. Each column adds to its total. $8 + 22 = 30$, $8 + 12 = 20$.

**Venn diagram:** the overlap is "both" (8). Outside the circles is "neither" (18).

**Relative frequency table.** Divide every count by the **grand total**. This gives the percent of **all** students.

| | **Job** | **No job** | **Total** |
|---|---|---|---|
| **Sport** | $\frac{8}{60} \approx 13\%$ | $\frac{22}{60} \approx 37\%$ | $\frac{30}{60} = 50\%$ |
| **No sport** | $\frac{12}{60} = 20\%$ | $\frac{18}{60} = 30\%$ | $\frac{30}{60} = 50\%$ |
| **Total** | $\frac{20}{60} \approx 33\%$ | $\frac{40}{60} \approx 67\%$ | 100% |

About 13% of **all** the students play a sport and have a job. The four inside cells add to 100%.

Work with a partner. Compare answers when you finish each problem.

1. 80 students said how they get to school.

| | **Bus** | **Car** | **Walk** | **Total** |
|---|---|---|---|---|
| **9th grade** | 18 | 12 | 10 | 40 |
| **10th grade** | 14 | 20 | 6 | 40 |
| **Total** | 32 | 32 | 16 | 80 |

   (a) How many 10th graders walk?

   (b) How many students ride the bus?

   (c) How many students are 9th graders who come by car?

   (d) How many students do **not** walk?

2. Fill in the missing numbers. Students said if they ate breakfast today.

| | **Yes** | **No** | **Total** |
|---|---|---|---|
| **11th grade** | 15 | | 35 |
| **12th grade** | | 10 | |
| **Total** | 27 | | 57 |

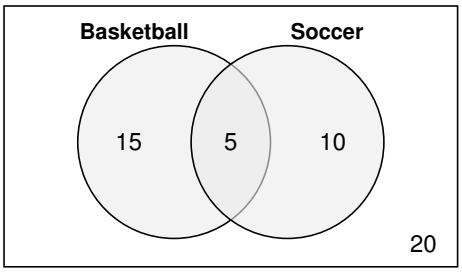
3. 50 people were asked about pets. 30 like cats. 26 like dogs. 12 like both.

   (a) Fill in the two-way table.

| | **Likes dogs** | **No dogs** | **Total** |
|---|---|---|---|
| **Likes cats** | | | |
| **No cats** | | | |
| **Total** | | | |

   (b) How many people like neither cats nor dogs?

4. This Venn diagram shows which sports 50 students play.




Make a two-way table from the Venn diagram.

| | **Soccer** | **No soccer** | **Total** |
|---|---|---|---|
| **Basketball** | | | |
| **No basketball** | | | |
| **Total** | | | |

5. Make a relative frequency table for problem 1. Divide each count by 80. Write percents.

| | **Bus** | **Car** | **Walk** | **Total** |
|---|---|---|---|---|
| **9th grade** | | | | |
| **10th grade** | | | | |
| **Total** | | | | |

(a) What percent of all the students are 10th graders who come by car?

(b) What percent of all the students walk?